SQL Assignment 2

SQL ASSIGNMENT 2

①
```
Select DISTINCT(department)
FROM students;
```

department
IT
HR
FINANCE

②
```
SELECT department,
       AVG(age) AS avg_age
FROM students
GROUP BY department;
```

department	avg_age
IT	20.5
HR	22
FINANCE	23

③
```
SELECT department,
       COUNT(student_id) AS student_count
FROM students
GROUP BY department
HAVING student_count > 1;
```

department	student_count
IT	2
HR	2

(4)
```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE age BETWEEN 21 AND 23;
```

student_id	name	age	department
2	Bob	22	HR
3	Charlie	21	IT
4	Diana	23	FINANCE
5	Eve	22	HR

(5)
```
SELECT student_id,
       name,
       age,
       department
FROM students
WHERE department = 'IT'
   OR department = 'HR'
AND age > 21;
```

student_id	name	age	department
2	Bob	22	HR
~~4~~	~~Diana~~	~~23~~	~~Finance~~
5	Eve	22	HR

⑥
```
SELECT department,
       SUM(credits) AS total_credits
FROM courses
GROUP BY department
HAVING total_credits > 5;
```

department	total_credits
IT	11

⑦
```
SELECT course_id,
       course_name,
       department,
       credits
FROM courses
WHERE credits <> 4;
```

course_id	course name	department	credits
101	SQl basics	IT	3
104	Excel	Finance	2
105	Statistics	HR	3

⑧
```
SELECT course_id,
       course_name,
       credits
FROM courses
ORDER BY credits DESC
Limit 3;
```

Course_id	course_name	credits
102	Python	4
103	Data Science	4
101	SQL basics	3

(9)
```
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM enrollments;
```

Max_grade	min_grade	avg_grade
90	78	84.6

(10)
```
SELECT carse_id,
       COUNT(enrollment_id) AS enrollment_count
FROM enrollments
GROUP BY course_id;
```

course_id	enrollment_count
101	1
102	1
103	1
104	1
105	1

(11)
```
SELECT department,
       SUM(salary) AS total_salary,
       SUM (bonus) AS total_bonus
FROM salaries
GROUP By department;
```

depaAment	total_salary	total_bonus
IT	122000	10500
HR	109 000	7500
FINANCE	70000	6000

(12)
```
SELECT   department,
         AVG(salary) AS avg_salary
FROM salaries
GROUP BY department
HAVING   avg_salary > 55000;
```

department	avg_salary
IT	61000
Finance	70000

(13)
```
SELECT   employee_id,
         name,
         salary,
         bonus,
         salary + bonus AS total_compensation
FROM salaries
WHERE  total_compensation > 60000;
```

employee_id	name	salary	bonus	total_compensation
1	Tom	60000	5000	65000
3	Spike	70000	6000	76000
4	Tyke	62000	5500	67500

(14)
```
SELECT   department,
         SUM(budget) AS total_budget
         AVG(budget) AS avg_budget
FROM projects
GROUP BY department
HAVING   avg_budget > 70000;
```

department	total_budget	avg_budget
IT	270000	135000
Finance	80000	80000

15

```
Select project_id,
       project_name,
       department,
       budget
FROM projects
Where department <> 'Marketing'
AND   budget Between 50000 AND 120000;
```

project_id	project_name	depatment	budget
1	AI App	IT	120 000
2	Payroll System	Finance	80 000
5	HR Portal	HR	60 000